

Osprey — Engineering Constraints

What this is. The boundaries any Osprey design decision has to sit inside, each with its source and the consequence of crossing it. Rules, physics, the hardware already on the boat, and the geometry of the hull. The rudder section at the end works the constraints through to a spec, as the worked example.

Sources. [R] the PEP26 Autonomy Division rules · [T §x] the Astudillo & Lee thesis, the as-built record · [SIM] the simulation in this repo · [TEAM] stated by the current team. Anything unverified says so.

1. Competition constraints [R]

Hard limits. A craft outside any of these does not race.

ID	CONSTRAINT	VALUE	OSPREY	CONSEQUENCE
C1	Battery total voltage	$\leq 55.5\text{ V}$	44.4 V nominal, 50.4 V full	Fixes the bus. Nothing higher is legal.
C2	Battery capacity	$< 500\text{ Ah}$	20.8 Ah	Not binding
C3	Payload	30 lb, removable, weighed dockside	13.6 kg in the model	Part of the loaded mass everything is sized to
C4	Race distance	2 statute miles	—	Sets energy and thermal duration
C5	Heat window	55 min	Sim finishes in ~7 min	Not binding on speed
C6	Kill on navigation loss	Must engage when GPS is missing or corrupted, demonstrated to judges	Not implemented	A required autonomy feature, not optional
C7	Positive buoyancy	Must float fully flooded	Verify	Constrains hull volume budget
C8	Tow bridle	Across both hulls, boat-hook reachable	Verify	
C9	Main disconnect + fuse	All propulsion current through one disconnect; fuse for 24–55 V	Present per thesis	
C10	Launch	Ramp to open water in 5 min	—	Constrains startup sequence and pre-arm checks
C11	No pre-built motor or kit	Motor, ESC, shaft, prop assembled by the team	Compliant	

Scoring shapes priorities. 10 pts per completed half-mile (40 total) against 20 / 16 / 12 / 8 / 4 for placing.

Finishing is worth twice winning. The required average speed to finish inside the heat is under 1 m/s. [R]

2. Physical limits — where the models stop applying

These are not design choices. They are places the physics changes regime.

ID	CONSTRAINT	VALUE	SOURCE	CONSEQUENCE
P1	Rudder cavitation ceiling	$\sigma < 0.5$ above 19.8 m/s (44 mph) ; $\sigma = 0.15$ at 80 mph	[SIM]	Attached-flow rudder model is void above ~44 mph. Anything above that is extrapolation. A conventional blade at 70–80 mph needs a ventilated or supercavitating section, not a bigger one.
P2	Ventilation onset	~10° deflection (swept 6–15°), latched, re-wets below ~6°	[SIM]	Yaw moment falls past onset (64 → 39 N·m at 30 mph). Useful deflection is ~10°, not the 35° the linkage reaches.
P3	Hull regime change	Displacement → planing over ~6–10 m/s; wetted area collapses 12×	[SIM]	Hull damping and rudder authority both change character. One gain set cannot span it.
P4	Manoeuvring-coefficient validity	Standard regressions fitted at $F_n < 0.3$; Osprey runs F_n 2.9–7.8	[SIM]	Hull derivatives are unknown to $\pm 5\times$. Every conclusion must survive that.
P5	Turn radius is speed-independent	R depends on deflection, not speed (u^2 authority ÷ u damping ÷ u required rate)	[SIM]	High speed never sizes the rudder. Low speed plus disturbance does.
P6	Speed sag in a hard turn	14–34% at full rudder	[SIM]	Gain schedule chases a moving operating point mid-turn. Limit commanded yaw rate.

3. Hardware constraints — what is already on the boat

ID	ITEM	LIMIT	WHERE THE BOAT SITS	SOURCE
H1	ESC Hydra Cobra 5 HV	180 A continuous, 9 kW at 50.4 V; requires 50 C packs	45 A at 30 mph (25%). 80 mph needs 180 A (100%)	[T §3.2] [SIM]
H2	Motor Castle 2535 680Kv	23,150 rpm logged	Under-loaded: 45 A where 100 was expected	[T §6]
H3	Battery 8 × Lectron Pro 22.2 V 5200 mAh 100 C, 2S-2P per ESC	100 C discharge; 924 Wh total	17 C at 80 mph. Energy is not a constraint at any speed (54% of usable for 2 mi at 80 mph)	[T §6.2] [SIM]
H4	Steering servo AGFRC A81FHM HV	74 kg·cm stall at 8.4 V	See §5 — depends entirely on the rudder stock position	[T §3.4.1]
H5	Cooling	Motor/ESC limit 82 °C	Sits at 40 °C — but raw-water side has never flowed and the boat has never run a race duration	[T §7.2]

ID	ITEM	LIMIT	WHERE THE BOAT SITS	SOURCE
H6	Driveline	Flex shaft retained by collet clamping force only	Failed at PEP26 by collet wear	[T §6.2]
H7	Prop Graupner 76 mm	Zero-thrust advance ratio unknown	Jo $\geq$ 1.22 needed for 80 mph to be reachable at all	[SIM]

Two derived constraints that follow:

- **H1 and H5 are the same constraint.** 180 A continuous assumes the cooling works. It does not. Nothing near the ESC rating is available until raw-water flow is proven.
- **H2 bounds speed, not H1.** The boat draws a quarter of what the ESC can supply. The limit on speed today is the propeller's loading of the motor. A larger ESC changes nothing.

4. Geometric constraints [T §3]

ID	CONSTRAINT	VALUE
G1	Length overall	7 ft 0 in (2.134 m)
G2	Deck beam / tunnel width	30.5 in / 18.0 in
G3	Loaded mass	97.4 lb (44.2 kg) with payload; 67.4 lb dry
G4	CG	~30% LOA forward of transom; battery rails slide 25–35%
G5	Prop lateral separation	~0.598 m (y_p = 0.299 m) — verify from CAD
G6	Rudder mount cannot sit below the bottom of the transom	"at high speeds it would cause too much drag, and at low speeds the mount would be dragging through the water" [T §3.4.2]
G7	Rudder mounts "as low as possible" within G6	The previous team's own concern was insufficient blade engagement [T §3.4.2]
G8	Rudder block sits atop the blade to meet the cable height	Longer blade extends down ; the block and cable geometry stay [T §3.4.2]

5. Rudder — constraints worked through to a spec

5.1 What is on the boat

MHZ Mystic C5000 rudder, chosen because it is made for the same catamaran at a similar scale [T §3.4.1]. Blade 150 mm × 30 mm. The thesis's own sizing assumed "**submerged blade area of 50% of 150 mm by 30 mm**" — i.e. about 3 in in the water — and the team's current estimate agrees [TEAM]. Still an estimate; a photograph of the transom on plane would settle it.



The MHZ Mystic C5000 rudder as supplied — blade and bracket. Fig 3.18a of the thesis.



As installed: the 3D-printed rudder mount on the transom, with the rudder block on top of the blade and the steering cable actuators. Fig 5.6 of the thesis.

The mount was designed to put the blade as low as G6 allows. The previous team wrote that *"there were concerns that the rudder we had was not long enough to get enough engagement with the water"* [T §3.4.2] . More depth was already on their list.

5.2 The constraints that shape the rudder

ID	CONSTRAINT	WHY
R1	Size on span, not area	Authority is $A_r \cdot CL_a$. Growing area at fixed span grows the chord, which collapses aspect ratio and cancels the gain: 30× the area buys 32% ; growing span instead buys 5× for a third of the area. [SIM]
R2	Chord stays at 30 mm	Follows from R1. Widening it is the one thing that does not help.
R3	Useful deflection is ~10° (P2)	The allocator clamps there. Authority past onset is not real.
R4	Stock at 20–25% chord aft of the leading edge	Hinge moment is $F \cdot (x_{cp} - x_{stock}) \cdot c$. At the CoP it is near zero; at the leading edge the lever is 0.25 c. This decides whether H4 is adequate — see 5.4.
R5	Mount stays put; blade extends downward (G6–G8)	The block-and-cable geometry at the top of the blade is unchanged.
R6	Same thin section, no end plate	A fatter section ventilates earlier on a surface-piercing blade. An end plate does nothing once the blade ventilates.

5.3 The spec — 3 in + 2 in [TEAM]

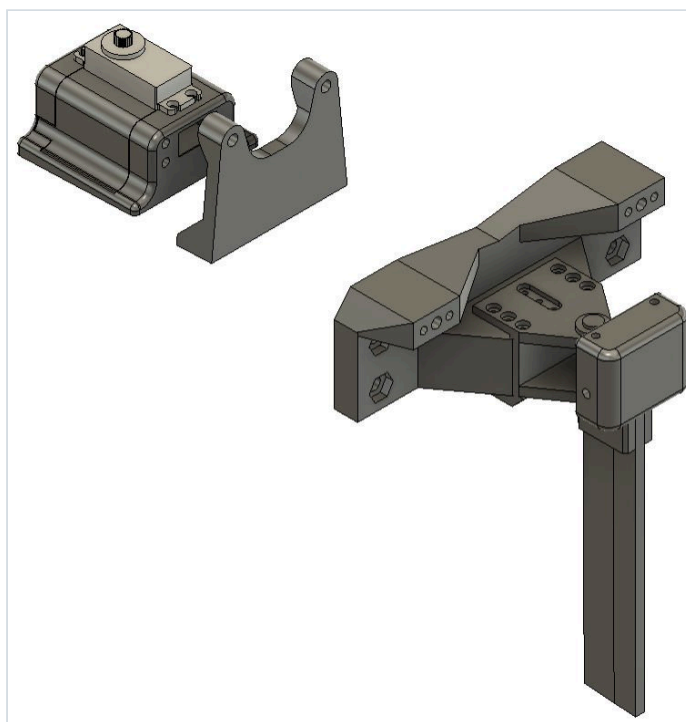
	CURRENT	NEW
Submerged span	3.0 in / 76 mm	5.0 in / 127 mm
Chord	1.18 in / 30 mm	1.18 in / 30 mm
Wetted area	3.5 in ²	5.9 in²
Aspect ratio	2.5	4.2
Total blade length	5.9 in	~8 in (same mount, 2 in longer below)
Stock position	unknown — measure	20–25% chord aft of LE
Max mechanical deflection	±35° assumed — not stated in the thesis; set by the pull-pull horn and mount tabs. Measure on the boat.	same linkage, same stop
Max useful deflection	±10° — ventilation onset (P2)	±10°. Limit the autopilot here, not at the stop. Past onset the blade gives <i>less</i> moment, not more.
Servo	AGFRC A81FHM HV, 74 kg·cm stall at 8.4 V , BEC-fed from one ESC [T §3.4.1]	Same servo is sufficient provided the stock is at 20–25% c — see 5.4. At a leading-edge stock it is exceeded above ~55 mph.
Authority $A_r \cdot CL_a$	1.0×	2.2×
Steady turn radius, full useful rudder	~20 m (9.5 LOA)	~11.5 m (5.4 LOA)

	CURRENT	NEW
Rudder drag at 10 m/s, full deflection	2.5% of hull	~4% of hull

Why 10° and not 35°. The linkage can reach 35°, but the blade ventilates at about 10° and the yaw moment then *falls* — 64 N·m at 10° to 39 N·m at 12° at 30 mph, latched until the blade unloads below ~6° [SIM] . Commanding past onset hands the controller a negative plant gain. The autopilot's steering output should be scaled so that full stick is $\pm 10^\circ$; the remaining travel is reserve for a manual recovery, not something the loop should ever see.

Going from aspect ratio 2.5 to 4.2 is the steepest part of the lift-slope curve, which is why 2 in of span buys 2.2× rather than the 1.7× the area ratio alone would suggest.

What the course asks for. Holding the 128 m oval needs under 2° of steady rudder at any speed [SIM] . The new blade is not for turning. It is for disturbance-rejection margin: across the $\pm 5\times$ parameter uncertainty the current blade spends 41% of a lap pinned at its useful limit; a blade in this class does not saturate [SIM] .



The full steering assembly in CAD — servo mount forward, cable routers, rudder mount aft. Fig 3.23 of the thesis.

5.4 Servo check against H4

Peak hinge moment occurs **at ventilation onset (~10°)**, where lift is highest, not at full deflection. For the 5 in blade against the 74 kg·cm servo:

SPEED	STOCK AT LE	STOCK AT 15% C	STOCK AT 20–25% C
30 mph	20 kg·cm (27%)	8 (11%)	~0
45 mph	45 (60%)	18 (24%)	~0
70 mph*	109 (148%)	44 (59%)	~0

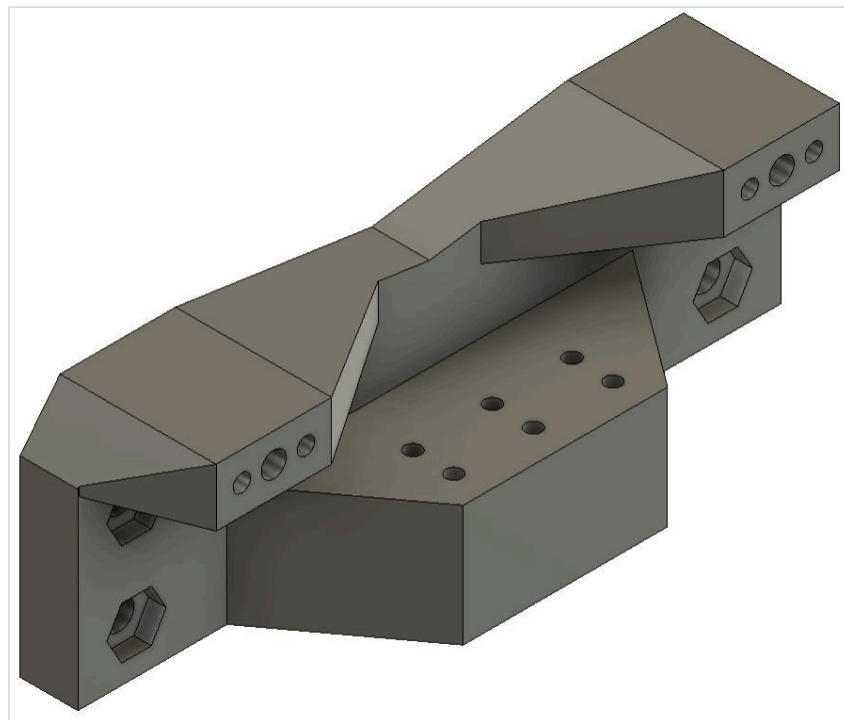
\ Beyond P1. Direction is certain, magnitude $\pm 30\%$.*

A leading-edge stock overloads the servo at 70 mph. Balanced at 20–25% chord, the existing servo covers every speed. R4 is therefore a hard constraint of this spec, not a preference. Ask the previous team where the current blade's stock sits before copying it.

5.5 Reconciling the thesis's 66 kg·cm

The thesis sized the servo as $F = \frac{1}{2}\rho V^2 \cdot A \cdot C_D$ with $C_D = 0.3$ at 80 mph, giving 436 N, then multiplied by the **1.5 cm servo horn arm** to get 66 kg·cm [T §3.4.1]. That lever is the force-to-torque conversion *at the servo*, not the hinge moment *at the stock*. The correct lever is $(x_{cp} - x_{stock}) \cdot c$, which for a balanced blade is near zero and for a leading-edge stock is 7.5 mm.

Re-run correctly at their own 80 mph condition: **45 kg·cm** worst case (LE stock), ~0 balanced [SIM]. Their figure was conservative by about 46% — which is why the 74 kg·cm servo has been adequate — and it would have stayed adequate at the leading edge only because the boat never went faster than 30 mph.



The rudder mount (Fig 3.21) and the rudder block that sits atop the blade to meet the cable height (Fig 3.22).



The servo and pull-pull cables installed in the hull. Fig 5.7 of the thesis.

5.6 What is still open on the rudder

	UNKNOWN	CLOSES BY
U1	Actual submerged depth on plane	One photo of the transom at 30 mph
U2	Current stock position on the blade	Callipers, or ask Monday (A2)
U3	Tiller arm radius — scales servo torque linearly	Callipers, or ask Monday (A3)
U4	Ventilation onset angle for this blade	Turn circles at 5° / 10° / 15° / 20°
U5	Draft on the trailer and at the ramp with a blade 2 in longer	Measure before the first launch
U6	Mechanical deflection limit of the linkage — the 35° in the model is assumed	Swing the rudder stop-to-stop with a protractor. Ten seconds.

6. Constraints that are still undefined

Things that will constrain the design once known, and currently do not because nobody has the number.

ID	UNKNOWN	WHAT IT WILL CONSTRAIN
X1	Speed target — 30, 50, 70 or 80 mph	ESC choice, wiring, cooling load, prop, and whether P1 matters. The decision to force first.
X2	Yaw inertia I_{zz}	Turn response; largest single unknown in the model ($\pm 50\%$). Bifilar pendulum, one hour.
X3	Hull derivatives	Everything in the control design. On-water zig-zags.

ID	UNKNOWN	WHAT IT WILL CONSTRAIN
X4	Prop $\kappa T(J)$	Whether any speed above ~ 26 m/s is reachable at all (H7).
X5	Course mark spacing vs lap length	Marks 0.25 mi apart put 0.5 mi of straight in a lap before turning; a lap cannot also be 0.5 mi. One of the two numbers is nominal.
X6	Steering allocation architecture	ArduPilot drives a rudder <i>or</i> twin motors, not both. Decide before tuning.